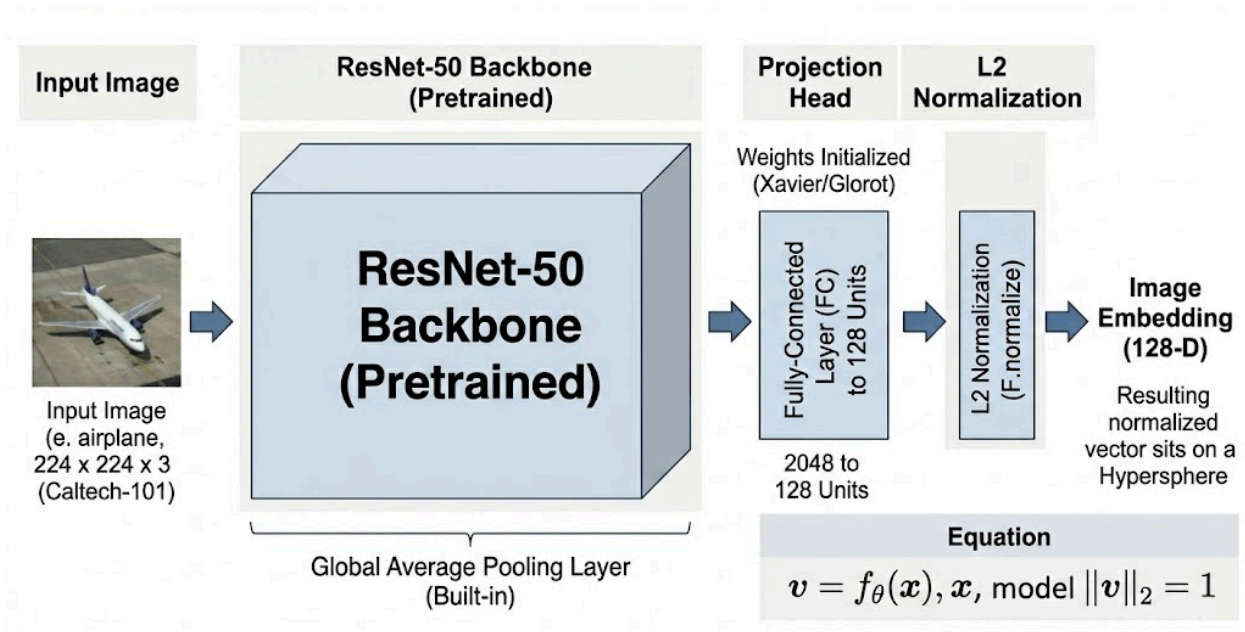


BSCS22030
SYED MUHAMMAD MURTAZA
DEEP LEARNING
ASSIGNMENT 3 REPORT

Model Architecture:



The architecture uses pretrained resnet50 as a backbone followed by a FC layer and L2 norm to output the 128 Dimensions embedding for a given image of size 224x224x3 as expected by the resnet50.

Throughout all experiments, the following hyperparamters were used:

- Batch size : 32
- Learning rate: 0.001
- Epochs : 15
- Optimizer : Adam's

Data Sampling Strategy:

The data sampling strategy used is dependent on the type of experiment.

For Contrastive Dataset:

We choose an image randomly. With probability of 0.5, we choose another image from same class and assign label 1, otherwise we choose a image from a different class and assign label 0

For Triplet Dataset:

We choose an image randomly(anchor). We choose one image from the same class that is not the anchor, this becomes our positive image. For negative image, we chose an image randomly from any other class.

For Hard Negative Sampling:

The sampling strategy lies in how we select the batch of images. We select images from 8 different classes, each class contributing 4 different images. In this way we ensure there is always a positive image available for any given image.

Loss Functions:

MATHEMATICAL FORMULATION OF LOSS FUNCTIONS

1. CONTRASTIVE LOSS (Experiment 1)

Calculated over a pair of image embeddings (v_1, v_2) and a binary label y , where $y = 1$ for positive pairs (same class) and $y = 0$ for negative pairs (different class).

$$L_{cont}(v_1, v_2, y) = y \cdot d(v_1, v_2)^2 + (1 - y) \cdot \max(0, m - d(v_1, v_2))^2$$

Where $d(v_1, v_2) = \|v_1 - v_2\|_2$ is the Euclidean distance and m is the margin.

Total loss over a batch of size B : $L_{Total} = \frac{1}{B} \sum_{i=1}^B L_{cont}^{(i)}$.

2. TRIPLET LOSS (Experiments 2 & 3)

Calculated over a triplet (a, p, n) consisting of an Anchor (a), a Positive (p), and a Negative (n).

$$L_{trip}(a, p, n) = \max(0, d(a, p) - d(a, n) + m)$$

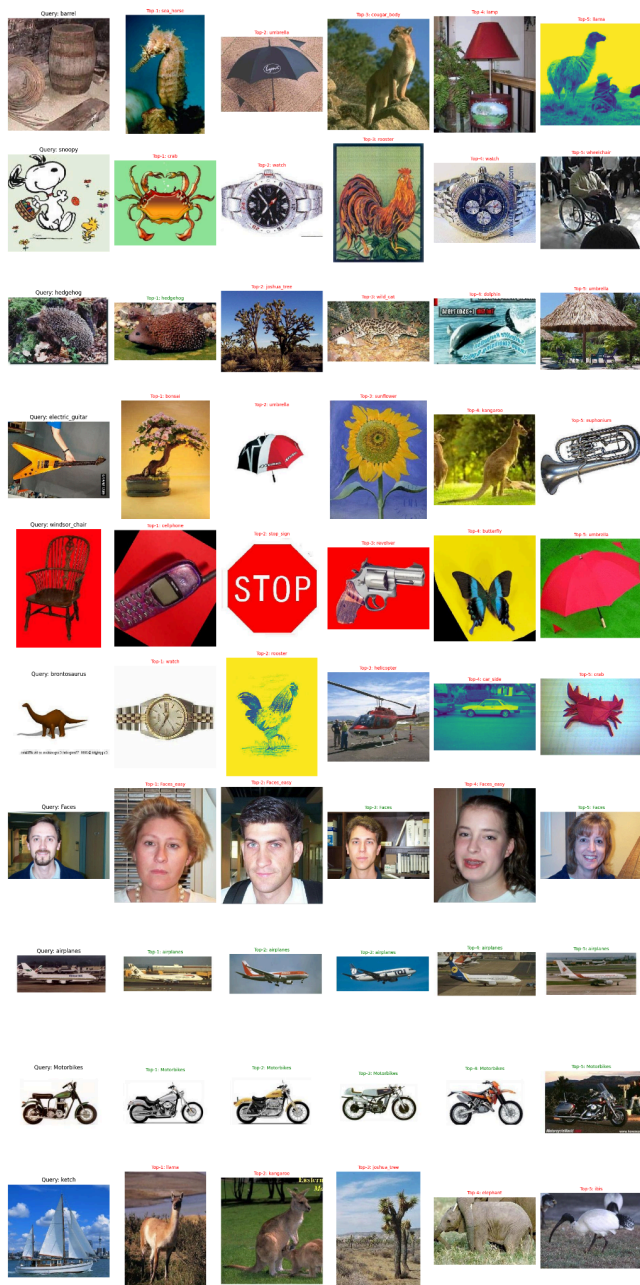
Where $d(x, y) = \|x - y\|_2$ and m is the margin. Experiment 3 utilizes Batch Hard Mining to select the samples.

Total loss over a batch of triplets: $L_{Total} = \frac{1}{|T|} \sum_T L_{trip}$, where $|T|$ is the number of triplets in the batch.

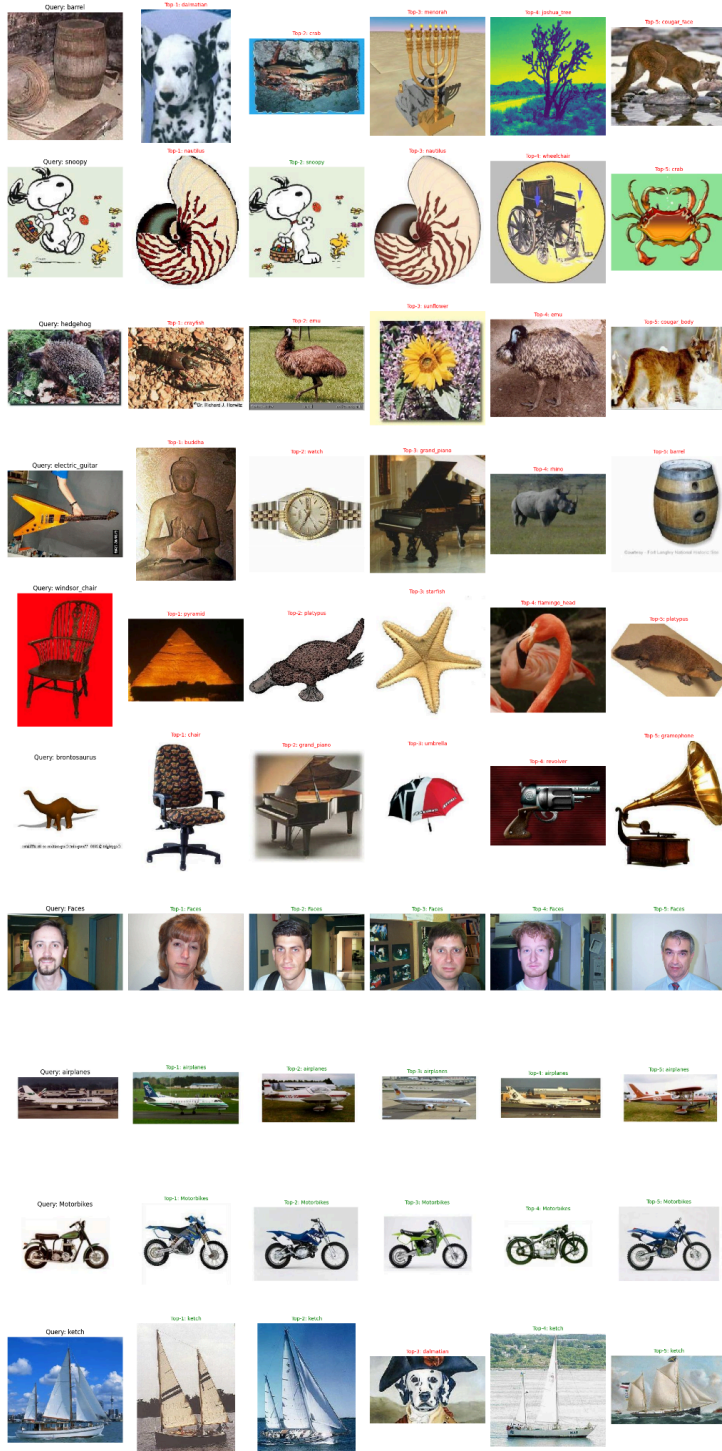
Retrieval Performance:

Model	recall@1	recall@5
Contrastive	0.3328	0.4820
Triplet	0.3497	0.5027
Hard Mining	0.6419	0.8171

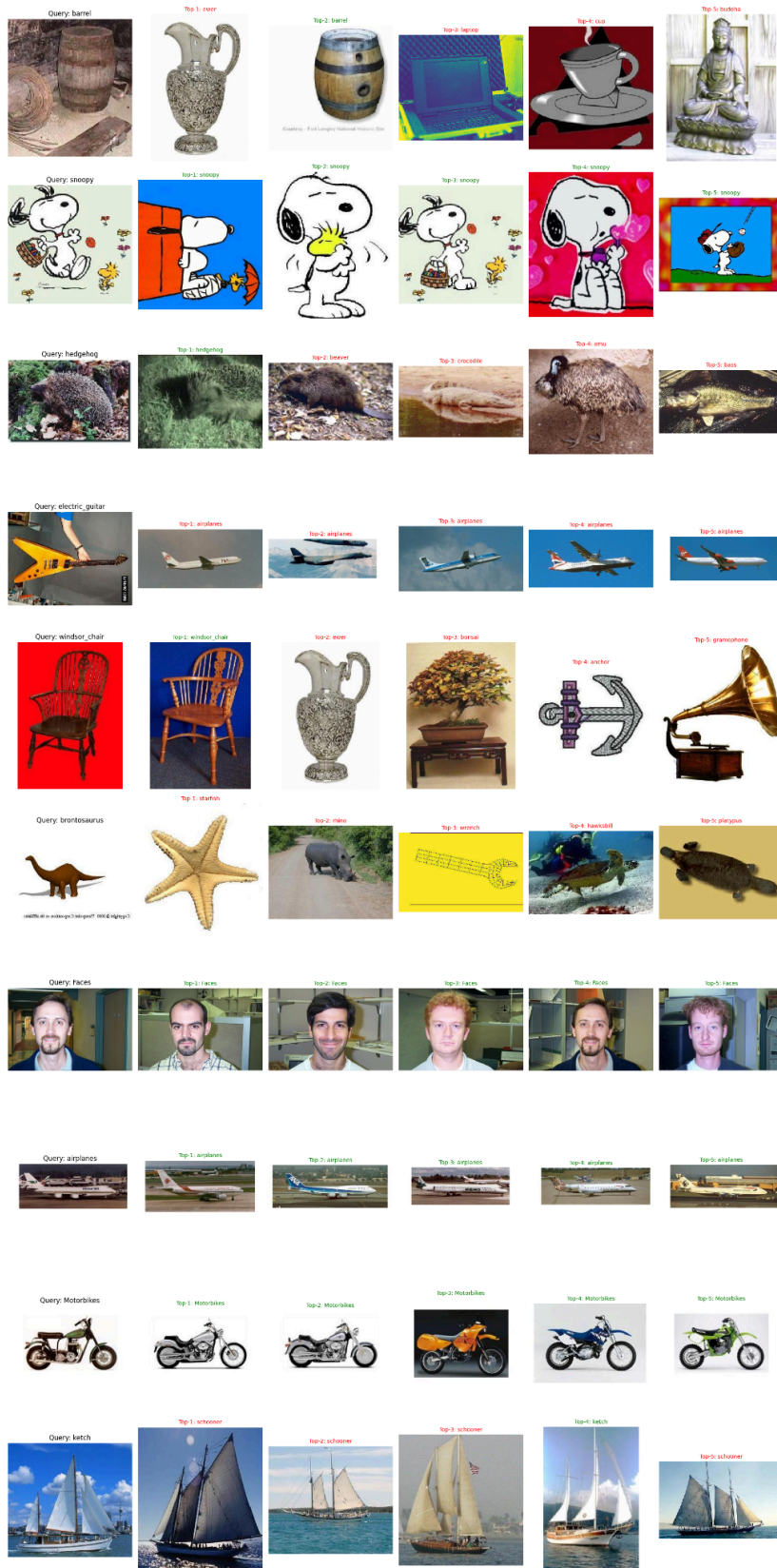
Retrieval Visualization: Contrastive:



Triplet:

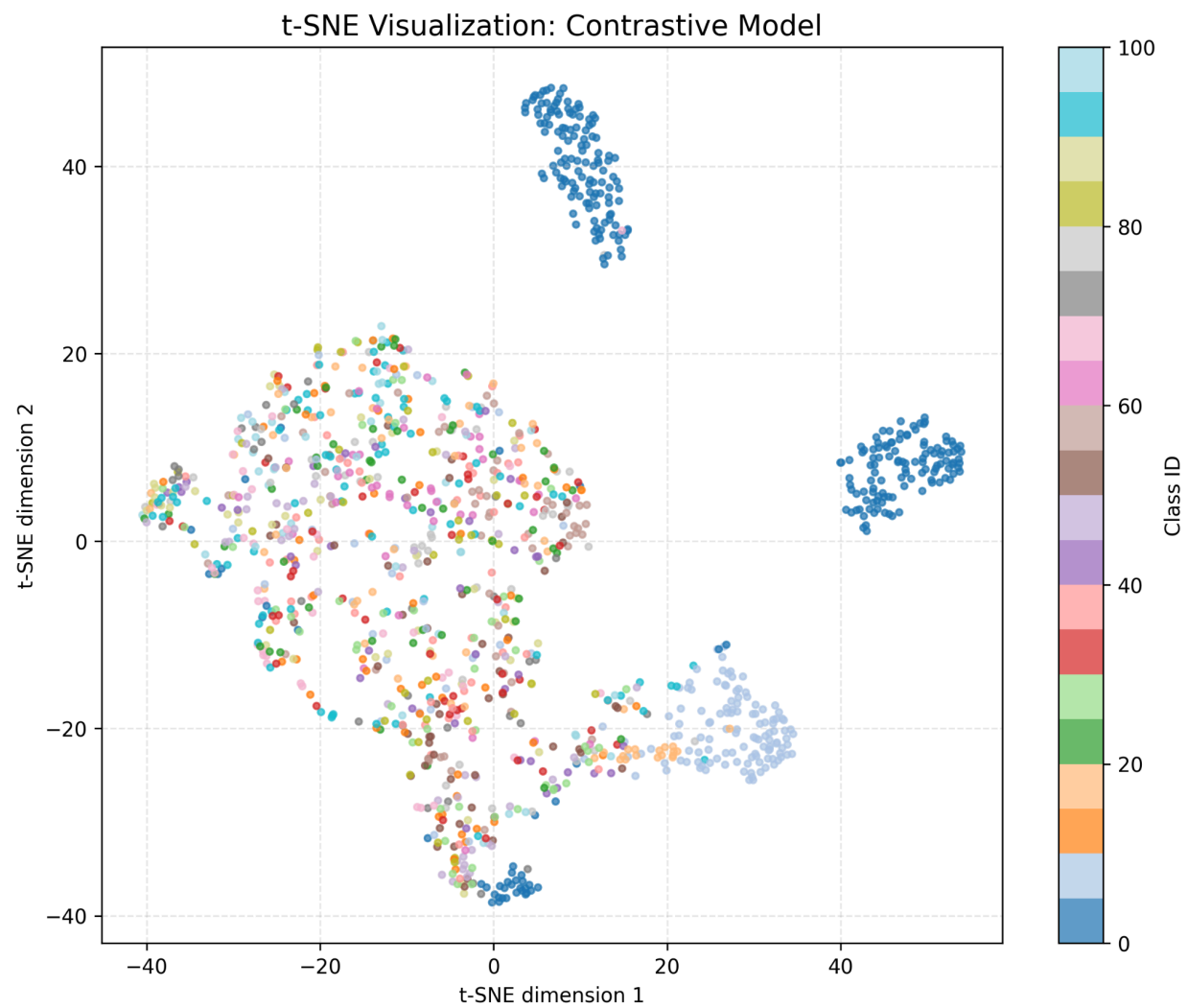


Hard Mining:

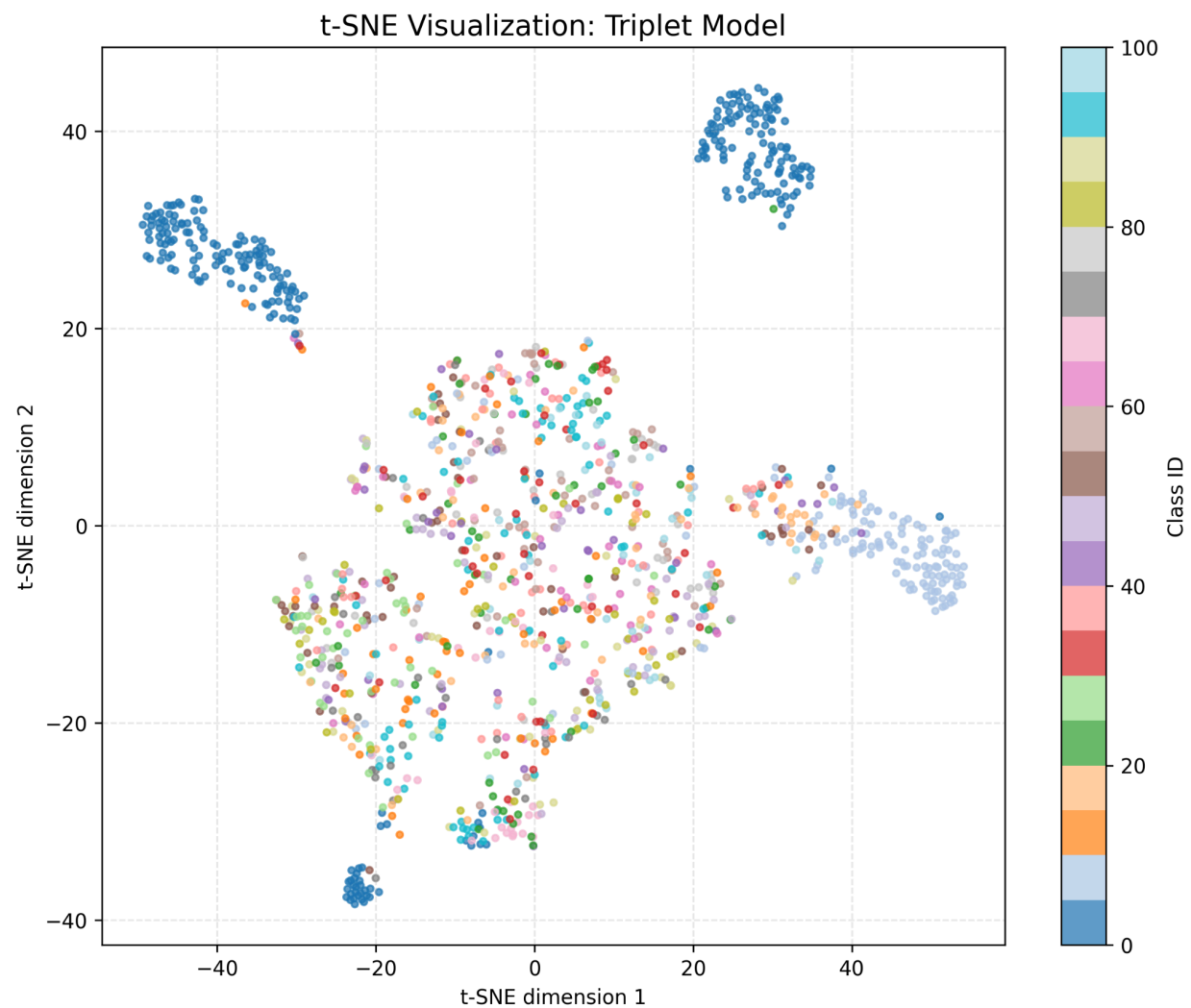


T-SNE Plots:

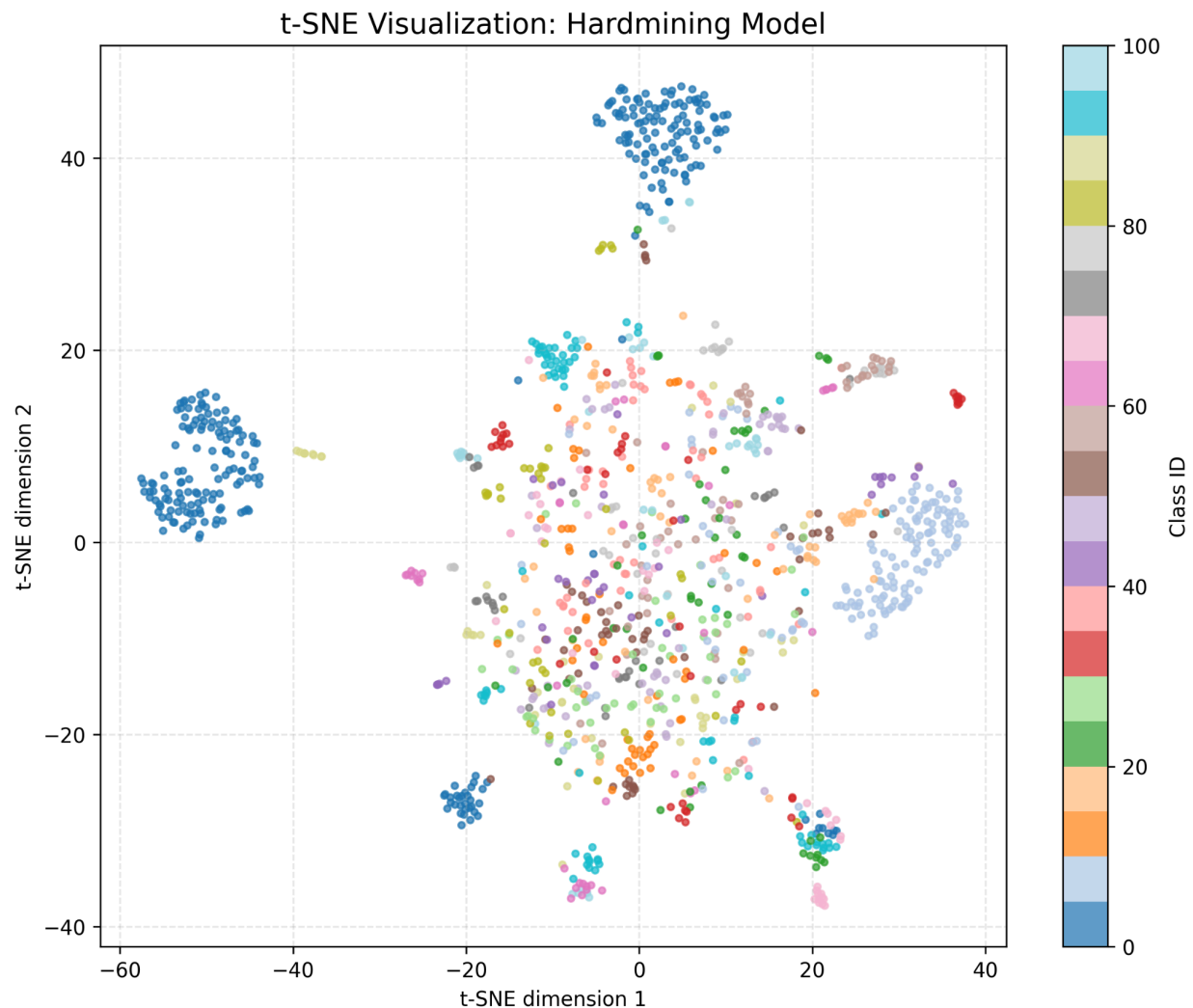
Contrastive:



Triplet:



Hard Mining:



Discussion:

It was an interesting assignment and i have much to say about it. The contrastive dataset and the triplet dataset each show visual progression in training epoch by epoch if we look at the training plots, however, in performance, it performs worse than hard mining as can be seen by the recall scores or the visualizations. Hard Mining clearly takes the edge despite the fact that visually in training, it seems stagnant. It is because when trained on hard examples, even an apparently poor learning would actually perform better.

Having said that, our model is still challenged alot by caltect-101 dataset. There are improvements to be made in each aspect. The most important

one right now is to force batch hard mining to learn more. To reduce the loss during training because if we are able to achieve that, we will achieve higher accuracies.

A failure case worth noticing is that when i was training hard mining, it failed when the sampler was sampling the images randomly. Because my batch size was 32, and there are about 101 classes, the probability of images of same classes appearing in one batch was low, due to which the only image closest to the anchor image, in case when there was a single image from a given class, was the image itself. The distance between them was 0, and it resulted in gradients being Nan. So i had to implement a custom sampler, which samples 32 images, total 8 classes and 4 images per class.

LLM Chat Link:

<https://gemini.google.com/share/73a1799cd260>